Structured Diophantine Identities and Universal Polynomial Parametrizations for Alternating Sums of Fourth Powers

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November 7, 2026

Abstract

We investigate structured quartic Diophantine identities of the form

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = n \cdot (b_1^4 - b_2^4 + b_3^4 - b_4^4)^2,$$

where $n \in \mathbb{Z}_{>0}$ is a fixed positive integer and $(a_1, a_2, a_3, a_4), (b_1, b_2, b_3, b_4) \in \mathbb{Z}^4$ are integer 4-tuples subject to four strict arithmetic non-degeneracy conditions: (i) all eight integers are non-zero; (ii) all eight integers are pairwise distinct; (iii) no integer is the negative of another (equivalently, $|a_1|, |a_2|, |a_3|, |a_4|, |b_1|, |b_2|, |b_3|, |b_4|$ are eight pairwise distinct positive integers); and (iv) $\gcd(a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4) = 1$.

In the first part of the paper, we resolve the benchmark case $n = 6$ by constructing two explicit, unscaled two-variable homogeneous polynomial parametrizations in $\mathbb{Z}[s, t]$ and $\mathbb{Z}[u, w]$ of degree $(4, 2)$, driven by the degenerate biquadratic conic $3X^2 = 2UV$. In particular, Family A produces infinitely many integer solutions for $n = 6$ in which both 4-tuples are individually primitive ($\gcd(a_1, a_2, a_3, a_4) = \gcd(b_1, b_2, b_3, b_4) = 1$), while Family B provides a low-coefficient parametrization satisfying $\gcd(b_1, b_2, b_3, b_4) = 1$ and $\gcd(a_1, a_2, a_3, a_4) = 2$. In the second part, we establish a Universal Quadric Lemma for the ternary form $3X^2 + U^2 + V^2 = W^2$ and compose it in two stages to obtain a closed-form polynomial parametrization in $\mathbb{Z}[n, p, q, s, t]$ valid for every fixed positive integer $n \geq 1$. Complete algebraic and number-theoretic proofs of the polynomial identities, individual and joint coprimality conditions, non-vanishing, and absolute-value distinctness are provided, together with explicit multi-parameter families and exact computational tables across $1 \leq n \leq 10$.

2020 Mathematics Subject Classification: Primary 11D25; Secondary 11D41, 11E25.
Keywords: Diophantine identities, biquadrates, alternating sums of fourth powers, polynomial parametrization, ternary quadrics, primitive solutions.

1 Introduction and Problem Formulation

The study of Diophantine equations involving fourth powers (biquadrates) occupies a central place in classical and modern number theory. Following Euler's celebrated two-parameter parametrization of the equal-sum biquadratic equation $A^4 + B^4 = C^4 + D^4$ in 1772 (surveyed in Dickson [7] and Hardy and Wright [11]), geometric and algebraic methods for constructing rational and parametric solutions were developed by Dyer [8], Richmond [15] (who treated the general linear combination $ax^4 + by^4 + cz^4 + dw^4 = 0$ when $abcd$ is a square), and Swinnerton-Dyer [16]. Comprehensive computational and structural studies of equal sums of like powers were subsequently carried out by Lander, Parkin, and Selfridge [14], Lander [13], Brudno [2], Zajta [17], and Bernstein [1]. In 1988, Elkies [9] combined elliptic-curve fibrations with lattice basis reduction to construct infinitely many primitive counterexamples to Euler's sum-of-powers conjecture for $A^4 + B^4 + C^4 = D^4$ (see also Guy [10]). More recently, parametric families for weighted biquadratic equations of the form $A^4 + hB^4 = C^4 + hD^4$ have been investigated by Choudhry [3, 4] and Izadi and Nabardi [12] (for $h = 2$), while new parametric solutions of the classical equal-sum equation $x^4 + y^4 = z^4 + w^4$—including odd-degree families of degree $6k + 3$ and the first even-degree bivariate polynomial solutions—were obtained by Choudhry [5] and Choudhry and Shamsi Zargar [6].

Whereas the classical equation $a_1^4 - a_2^4 + a_3^4 - a_4^4 = 0$ seeks vanishing alternating sums of four biquadrates, a natural structural generalization asks when a non-vanishing alternating sum of four fourth powers can equal a prescribed integer multiple n of the square of another alternating sum of four fourth powers, subject to strict non-degeneracy and primitivity constraints that rule out trivial cancellations or common scale factors.

We formulate both the specific case $n = 6$ and the general case of an arbitrary fixed positive integer $n \in \mathbb{Z}_{>0}$ as follows.

Problem 1.1 (The Case $n = 6$ and General $n \in \mathbb{Z}_{>0}$)**.** Let $n \in \mathbb{Z}_{>0}$ be a fixed positive integer (specifically $n = 6$, and more generally any $n \geq 1$). Determine whether there exist infinitely many pairs of integer 4-tuples

$$(a_1, a_2, a_3, a_4) \in \mathbb{Z}^4, \qquad (b_1, b_2, b_3, b_4) \in \mathbb{Z}^4$$

and explicit polynomial parametrizations generating them such that:

(C1) Non-vanishing: All $a_i, b_j \in \mathbb{Z} \setminus \{0\}$ for $1 \leq i, j \leq 4$.

(C2) Pairwise distinctness: All eight integers $a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4$ are pairwise distinct.

(C3) No opposite pairs: No integer is the negative of another; that is, $a_i \neq -a_j$, $b_i \neq -b_j$, and $a_i \neq -b_j$ for all $1 \leq i, j \leq 4$. Notice that Conditions **(C1)–(C3)** together are equivalent to requiring that the eight absolute values

$$|a_1|, |a_2|, |a_3|, |a_4|, |b_1|, |b_2|, |b_3|, |b_4|$$

be pairwise distinct positive integers.

(C4) Joint primitivity: $\gcd(a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4) = 1$.

(C5) Diophantine identity: The following quartic-squared relation holds:

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = n \cdot \left(b_1^4 - b_2^4 + b_3^4 - b_4^4\right)^2. \tag{1}$$

Summary of Main Results: We prove that Problem 1.1 has an affirmative answer for every positive integer $n \geq 1$. For $n = 6$, we construct two explicit, unscaled two-variable homogeneous polynomial parametrizations of degree $(4, 2)$ in $\mathbb{Z}[s, t]$ and $\mathbb{Z}[u, w]$ (Theorems 3.2 and 3.3), the first of which achieves $\gcd(a_1, a_2, a_3, a_4) = \gcd(b_1, b_2, b_3, b_4) = 1$ simultaneously, while the second has low coefficients with $\gcd(b_1, b_2, b_3, b_4) = 1$ and $\gcd(a_1, a_2, a_3, a_4) = 2$. For an arbitrary positive integer $n \geq 1$, we establish the Universal Quadric Lemma (Lemma 2.2) and prove a universal two-stage polynomial theorem in $\mathbb{Z}[n, p, q, s, t]$ (Theorem 4.2) that yields infinitely many primitive, non-degenerate solutions for every $n \geq 1$ without requiring ad-hoc conic searches. We note two structural features of the general construction: (i) for fixed (n, p, q), Theorem 4.2 is a coprime scaling construction $a_i(s, t) = M(s, t)A_i(p, q)$ on top of a fixed base tuple (A_1, A_2, A_3, A_4) (so the ratios $a_1 : a_2 : a_3 : a_4$ depend on (n, p, q) rather than (s, t), though varying (p, q) varies the projectivized a-tuple); and (ii) there is a natural size trade-off—because Theorem 4.2 avoids factoring the stage-1 constant, its unreduced specializations have large entries (for $(p, q, s, t) = (2, 1, 6, 1)$ and $n = 6$, the a_i have 28 digits and the b_j have 15 digits), whereas the square-free reduction of Corollary 4.3 and the unscaled $n = 6$ families of Section 3 yield much smaller solutions.

2 Central Difference Identities and the Universal Quadric Lemma

Our systematic method begins by restricting the four variables of an alternating fourth-power sum to lie on the hyperplane $t_1 + t_3 = t_2 + t_4$, which transforms the quartic polynomial into a factored product of two linear parameters and a positive-definite ternary quadratic form.

Lemma 2.1 (Central Second-Difference Identity for Fourth Powers)**.** *Let R be any commutative ring, and let $X, U, V \in R$. Define the four linear forms*

$$t_1 = X + U + V, \qquad t_2 = X + U - V, \qquad t_3 = X - U - V, \qquad t_4 = X - U + V. \tag{2}$$

Then the following polynomial identity holds in $R[X, U, V]$:

$$t_1^4 - t_2^4 + t_3^4 - t_4^4 = 16\,UV\big(3X^2 + U^2 + V^2\big). \tag{3}$$

Furthermore, if $t_i' = (X \pm U \pm V)/2 \in R$ (when $X + U + V$ is divisible by 2), then

$$(t_1')^4 - (t_2')^4 + (t_3')^4 - (t_4')^4 = UV\big(3X^2 + U^2 + V^2\big). \tag{4}$$

Proof. Recall the first central difference of a fourth power: for any $y, V \in R$, expanding $(y + V)^4$ and $(y - V)^4$ by the binomial theorem yields

$$(y + V)^4 - (y - V)^4 = 2\big(4y^3V + 4yV^3\big) = 8Vy\big(y^2 + V^2\big).$$

Applying this identity first with $y = X + U$ (for $t_1^4 - t_2^4$) and then with $y = X - U$ (for $t_4^4 - t_3^4$), we obtain

$$\begin{aligned} t_1^4 - t_2^4 + t_3^4 - t_4^4 &= \big[(X + U + V)^4 - (X + U - V)^4\big] - \big[(X - U + V)^4 - (X - U - V)^4\big] \\ &= 8V\Big[(X + U)\big((X + U)^2 + V^2\big) - (X - U)\big((X - U)^2 + V^2\big)\Big] \\ &= 8V\Big[\big((X + U)^3 - (X - U)^3\big) + V^2\big((X + U) - (X - U)\big)\Big]. \end{aligned}$$

Since $(X + U)^3 - (X - U)^3 = 6X^2U + 2U^3$ and $(X + U) - (X - U) = 2U$, substituting these expressions gives

$$t_1^4 - t_2^4 + t_3^4 - t_4^4 = 8V\big(6X^2U + 2U^3 + 2UV^2\big) = 16\,UV\big(3X^2 + U^2 + V^2\big),$$

proving (3). Dividing each t_i by 2 divides the fourth powers by $2^4 = 16$, which immediately yields (4). □

To convert the right-hand side of (3) into a prescribed multiple of a square, we must make the ternary quadratic form $3X^2 + U^2 + V^2$ a perfect square W^2 while simultaneously controlling the product UV. The following lemma provides a two-parameter polynomial solution to $3X^2 + U^2 + V^2 = W^2$ in which V is itself a square and U is a monomial in the two parameters.

Lemma 2.2 (Universal Quadric Lemma)**.** *Let R be a commutative ring and let $u, r \in R$. Define*

$$X = r(u + 4r), \qquad U = 3ru, \qquad V = (u + r)^2, \qquad W = u^2 + 2ur + 7r^2. \tag{5}$$

Then:

(i) The ternary quadric relation $3X^2 + U^2 + V^2 = W^2$ holds identically in $R[u, r]$.

(ii) The linear forms t_1, t_2, t_3, t_4 defined by (2) admit the explicit factorizations

$$\begin{aligned} t_1 &= u^2 + 6ur + 5r^2 = (u + r)(u + 5r), \\ t_2 &= -u^2 + 2ur + 3r^2 = (3r - u)(u + r), \\ t_3 &= -u^2 - 4ur + 3r^2, \\ t_4 &= u^2 + 5r^2, \end{aligned} \tag{6}$$

and satisfy the biquadratic identity

$$t_1^4 - t_2^4 + t_3^4 - t_4^4 = 3ru \cdot \Big(4(u + r)\big(u^2 + 2ur + 7r^2\big)\Big)^2. \tag{7}$$

Proof. For (i), observe that $W - V = 6r^2$ and $W + V = 2(u^2 + 2ur + 4r^2)$, so the difference of squares $W^2 - V^2$ factors directly as

$$W^2 - V^2 = (W - V)(W + V) = 12r^2(u^2 + 2ur + 4r^2) = 3r^2(u + 4r)^2 + 9r^2u^2 = 3X^2 + U^2,$$

which rearranges immediately to $3X^2 + U^2 + V^2 = W^2$.

For (ii), direct substitution of (5) into (2) gives

$$\begin{aligned}
t_1 &= (ur + 4r^2) + 3ru + (u^2 + 2ur + r^2) = u^2 + 6ur + 5r^2 = (u + r)(u + 5r),\\
t_2 &= (ur + 4r^2) + 3ru - (u^2 + 2ur + r^2) = -u^2 + 2ur + 3r^2 = (3r - u)(u + r),\\
t_3 &= (ur + 4r^2) - 3ru - (u^2 + 2ur + r^2) = -u^2 - 4ur + 3r^2,\\
t_4 &= (ur + 4r^2) - 3ru + (u^2 + 2ur + r^2) = u^2 + 5r^2.
\end{aligned}$$

Finally, by Lemma 2.1 and part (i), we have

$$t_1^4 - t_2^4 + t_3^4 - t_4^4 = 16\,UVW^2 = 16(3ru)(u + r)^2W^2 = 3ru \cdot \big(4(u + r)W\big)^2,$$

which is (7). □

We now establish the exact arithmetic conditions on $(u, r) \in \mathbb{Z}^2$ under which the resulting 4-tuple (t_1, t_2, t_3, t_4) is primitive, non-zero, and has pairwise distinct absolute values.

Lemma 2.3 (Primitivity and Non-Degeneracy of the Quadric Tuple)**.** *Let $u, r \in \mathbb{Z}_{>0}$ be positive integers satisfying:*

(a) $\gcd(u, r) = 1$*;*

(b) $u \not\equiv r \pmod 2$ *(opposite parity); and*

(c) $u + r \not\equiv 0 \pmod 3$.

(Note that conditions (a) and (b) automatically imply $u \neq 3r$, since $u = 3r$ with $\gcd(u, r) = 1$ would force $(u, r) = (3, 1)$, contradicting opposite parity.) Then the four integers (t_1, t_2, t_3, t_4) defined by (6) satisfy $\gcd(t_1, t_2, t_3, t_4) = 1$, are all non-zero, and have pairwise distinct absolute values $|t_1|, |t_2|, |t_3|, |t_4|$.

Proof. First, we prove that $\gcd(t_1, t_2, t_3, t_4) = 1$. Let $g = \gcd(t_1, t_2, t_3, t_4)$. From the linear combinations

$$t_1 + t_2 + t_3 + t_4 = 4X = 4r(u + 4r),$$
$$(t_1 + t_2) - (t_3 + t_4) = 4U = 12ru, \qquad (t_1 + t_4) - (t_2 + t_3) = t_1 - t_2 - t_3 + t_4 = 4V = 4(u + r)^2,$$

any prime $p \mid g$ must divide $4r(u + 4r)$, $12ru$, and $4(u + r)^2$.

- *Case $p = 2$:* Since $u \not\equiv r \pmod 2$, $u + r$ is odd, so $V = (u + r)^2$ is odd, whereas $X + U = r(4u + 4r) = 4r(u + r)$ is divisible by 4, and $X - U = r(4r - 2u) = 2r(2r - u)$ is even. Hence every $t_i = (X \pm U) \pm V$ is odd, which proves $2 \nmid g$.
- *Case $p = 3$:* Since $u + r \not\equiv 0 \pmod 3$, $3 \nmid (u + r)$, so $3 \nmid 4(u + r)^2 = 4V$. Thus $3 \nmid g$.
- *Case $p > 3$:* Since $p \nmid 12$, p must divide both $(u + r)^2$ and ru. Hence $p \mid (u + r)$ and either $p \mid u$ or $p \mid r$. Either alternative implies $p \mid u$ and $p \mid r$, contradicting $\gcd(u, r) = 1$.

Thus $g = 1$.

Second, we verify non-vanishing ($t_i \neq 0$). Since $u, r > 0$, we have $t_1 = (u + r)(u + 5r) > 0$ and $t_4 = u^2 + 5r^2 > 0$. Moreover, $t_2 = (3r - u)(u + r) \neq 0$ because $u = 3r$ and $\gcd(u, r) = 1$

would force $(u, r) = (3, 1)$, which is excluded by hypothesis (b) ($u \not\equiv r \pmod 2$). Finally, $t_3 = -u^2 - 4ur + 3r^2 = 7r^2 - (u + 2r)^2 \neq 0$ because 7 is not a square in $\mathbb{Q}$.

Third, we verify that $|t_1|, |t_2|, |t_3|, |t_4|$ are pairwise distinct. For any two integers t_i, t_j, we have $|t_i| = |t_j|$ if and only if $t_i^2 - t_j^2 = (t_i - t_j)(t_i + t_j) = 0$, which holds if and only if $t_i - t_j = 0$ or $t_i + t_j = 0$. For all $1 \leq i < j \leq 4$, the twelve sums and differences $t_i \pm t_j$ belong to the set

$$\{\pm 2X, \pm 2U, \pm 2V, 2(X \pm U), 2(X \pm V), 2(U \pm V)\}.$$

Specifically:

- $t_1 - t_2 = t_4 - t_3 = 2V = 2(u + r)^2 > 0$ and $t_1 - t_4 = t_2 - t_3 = 2U = 6ru > 0$.
- $t_1 + t_3 = t_2 + t_4 = 2X = 2r(u + 4r) > 0$.
- $t_1 + t_2 = 2(X + U) = 8r(u + r) > 0$ and $t_3 + t_4 = 2(X - U) = 4r(2r - u) \neq 0$ (since $u = 2r$ and $\gcd(u, r) = 1$ force $(u, r) = (2, 1)$, which gives $u + r = 3$ and is excluded by hypothesis (c)).
- $t_1 + t_4 = 2(X + V) = 2(u^2 + 3ur + 5r^2) > 0$ and $t_2 + t_3 = 2(X - V) = 2(3r^2 - ur - u^2) \neq 0$ (as $u^2 + ur - 3r^2$ has non-square discriminant $13r^2$).
- $t_1 - t_3 = 2(U + V) = 2(u^2 + 5ur + r^2) > 0$ and $t_2 - t_4 = 2(U - V) = -2(u^2 - ur + r^2) < 0$.

Because none of the twelve quantities $t_i - t_j$ or $t_i + t_j$ ($1 \leq i < j \leq 4$) vanishes, we have $|t_i| \neq |t_j|$ for all $i \neq j$. □

3 Complete Resolution of the Case $n = 6$: Two Unscaled Polynomial Families

Before treating general n, we examine why $n = 6$ admits compact, unscaled degree-$(4, 2)$ polynomial parametrizations. Under the conic specialization $3X^2 = 2UV$, the ternary quadratic form becomes a square $3X^2 + U^2 + V^2 = (U + V)^2$, while the remaining factor in (3) is $16UV = 24X^2 = 6(2X)^2$—that is, 6 arises naturally as the square-free part of 24.

Proposition 3.1 (The Degenerate Conic $3X^2 = 2UV$ for $n = 6$)**.** *Suppose X, U, V satisfy the conic relation $3X^2 = 2UV$. Then:*

$$3X^2 + U^2 + V^2 = U^2 + 2UV + V^2 = (U + V)^2 \qquad \textit{and} \qquad 16\,UV = 24X^2 = 6(2X)^2.$$

Consequently, the linear forms $a_1 = X + U + V$, $a_2 = X + U - V$, $a_3 = X - U - V$, $a_4 = X - U + V$ satisfy

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = 6 \cdot \big(2X(U + V)\big)^2. \tag{8}$$

In particular, setting $U = 6P^2$, $V = Q^2$, and $X = 2PQ$ satisfies $3X^2 = 2UV$ identically and gives

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = 6 \cdot \Big(4PQ\big(6P^2 + Q^2\big)\Big)^2. \tag{9}$$

Proof. Substituting $3X^2 = 2UV$ and $16UV = 24X^2$ into Lemma 2.1 gives

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = 16\,UV(3X^2 + U^2 + V^2) = 24X^2(U + V)^2 = 6\big(2X(U + V)\big)^2.$$

When $U = 6P^2$, $V = Q^2$, and $X = 2PQ$, we have $3X^2 = 12P^2Q^2 = 2UV$ and $2X(U + V) = 4PQ(6P^2 + Q^2)$. □

To match $4PQ(6P^2 + Q^2)$ with $b_1^4 - b_2^4 + b_3^4 - b_4^4$, we represent (b_1, b_2, b_3, b_4) via the half-integer form (4) of Lemma 2.1 using auxiliary parameters (A_b, B_b, C_b):

$$b_1 = \frac{A_b + B_b + C_b}{2}, \qquad b_2 = \frac{A_b + B_b - C_b}{2}, \qquad b_3 = \frac{A_b - B_b - C_b}{2}, \qquad b_4 = \frac{A_b - B_b + C_b}{2},$$

so that $b_1^4 - b_2^4 + b_3^4 - b_4^4 = B_bC_b(3A_b^2 + B_b^2 + C_b^2)$. Setting $B_b = m_1P$ and $C_b = m_2Q$ equates the two sides precisely when

$$B_bC_b(3A_b^2 + B_b^2 + C_b^2) = 4PQ(6P^2 + Q^2) \iff 3m_1m_2A_b^2 = (24 - m_1^3m_2)P^2 + (4 - m_1m_2^3)Q^2. \quad (10)$$

Two natural choices of (m_1, m_2) produce rational conics that can be parametrized completely over $\mathbb{Z}[s, t]$ and $\mathbb{Z}[u, w]$.

3.1 Family A: Simultaneously Primitive Polynomials ($m_1 = 1, m_2 = 1$)

Setting $m_1 = 1$ and $m_2 = 1$ in (10) yields $3A_b^2 = 23P^2 + 3Q^2$. Since $3 \mid 23P^2$, we write $P = 3P_0$, which reduces the conic to the difference of squares

$$A_b^2 - Q^2 = (A_b - Q)(A_b + Q) = 69P_0^2.$$

Factoring $69P_0^2 = (6s^2)(46t^2)$ with $P_0 = 2st$ gives $A_b - Q = 6s^2$ and $A_b + Q = 46t^2$, whence

$$P = 6st, \qquad Q = 23t^2 - 3s^2, \qquad A_b = 23t^2 + 3s^2.$$

Theorem 3.2 (Family A for $n = 6$)**.** *For any integers $s, t \in \mathbb{Z}$, define the quadratic polynomials*

$$\begin{aligned} b_1(s,t) &= 23t^2 + 3st = t(23t + 3s), & b_2(s,t) &= 3s^2 + 3st = 3s(s + t), \\ b_3(s,t) &= 3s^2 - 3st = 3s(s - t), & b_4(s,t) &= 23t^2 - 3st = t(23t - 3s), \end{aligned} \quad (11)$$

and, setting $U = 216s^2t^2$, $V = (23t^2 - 3s^2)^2$, and $X = 12st(23t^2 - 3s^2)$, define the quartic polynomials

$$\begin{aligned} a_1(s,t) &= V + U + X = 9s^4 - 36s^3t + 78s^2t^2 + 276st^3 + 529t^4, \\ a_2(s,t) &= V - U - X = 9s^4 + 36s^3t - 354s^2t^2 - 276st^3 + 529t^4, \\ a_3(s,t) &= V + U - X = 9s^4 + 36s^3t + 78s^2t^2 - 276st^3 + 529t^4, \\ a_4(s,t) &= V - U + X = 9s^4 - 36s^3t - 354s^2t^2 + 276st^3 + 529t^4. \end{aligned} \quad (12)$$

Then:

(i) The quartic-squared identity

$$a_1(s,t)^4 - a_2(s,t)^4 + a_3(s,t)^4 - a_4(s,t)^4 = 6\big(b_1(s,t)^4 - b_2(s,t)^4 + b_3(s,t)^4 - b_4(s,t)^4\big)^2$$

holds identically in $\mathbb{Z}[s, t]$.

(ii) For every pair of integers $t > s \geq 1$ such that $\gcd(s, t) = 1$, $s \not\equiv t \pmod 2$, $3 \nmid t$, and $23 \nmid s$, the specialized tuples satisfy both individual primitivity conditions

$$\gcd(a_1, a_2, a_3, a_4) = 1 \qquad \textit{and} \qquad \gcd(b_1, b_2, b_3, b_4) = 1$$

*(and hence $\gcd(a_1, \ldots, b_4) = 1$), as well as Conditions **(C1)–(C3)** (all eight numbers are non-zero and have pairwise distinct absolute values). Moreover, whenever $10t \geq 11s$ (and in particular whenever $t \geq 2s$), the eight absolute values satisfy the explicit strict ordering*

$$0 < |b_3(s,t)| < b_2(s,t) < b_4(s,t) < b_1(s,t) < a_2(s,t) < a_3(s,t) < a_4(s,t) < a_1(s,t). \quad (13)$$

Proof. (i) By construction, $P = 6st$, $Q = 23t^2 - 3s^2$, and $A_b = 23t^2 + 3s^2$ satisfy

$$3A_b^2 + P^2 + Q^2 = 3(23t^2+3s^2)^2 + 36s^2t^2 + (23t^2-3s^2)^2 = 4\big(216s^2t^2 + (23t^2-3s^2)^2\big) = 4(6P^2+Q^2).$$

Moreover, $(A_b + P + Q)/2 = b_1$, $(A_b + P - Q)/2 = b_2$, $(A_b - P - Q)/2 = b_3$, and $(A_b - P + Q)/2 = b_4$. By Lemma 2.1,

$$b_1^4 - b_2^4 + b_3^4 - b_4^4 = PQ(3A_b^2 + P^2 + Q^2) = 4PQ(6P^2+Q^2).$$

Meanwhile, comparing (12) with the linear forms $t_1 = X + U + V$, $t_2 = X + U - V$, $t_3 = X - U - V$, $t_4 = X - U + V$ of Lemma 2.1, we have $(a_1, -a_2, -a_3, a_4) = (t_1, t_2, t_3, t_4)$ with $U = 6P^2$, $V = Q^2$, and $X = 2PQ$. Since $(-x)^4 = x^4$, Proposition 3.1 gives

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = t_1^4 - t_2^4 + t_3^4 - t_4^4 = 6 \cdot \big(4PQ(6P^2+Q^2)\big)^2 = 6 \cdot \big(b_1^4 - b_2^4 + b_3^4 - b_4^4\big)^2.$$

(ii) Assume $t > s \geq 1$ with $\gcd(s,t) = 1$, $s \not\equiv t \pmod 2$, $3 \nmid t$, and $23 \nmid s$. First, since s, t have opposite parity, $Q = 23t^2 - 3s^2$ is odd. Because $3 \nmid t$, $Q \equiv 2t^2 \not\equiv 0 \pmod 3$. Any prime $p \mid s$ cannot divide $23t^2$ (since $\gcd(s,t) = 1$ and $23 \nmid s$), and any prime $p \mid t$ cannot divide $3s^2$ (since $\gcd(s,t) = 1$ and $3 \nmid t$). Thus $\gcd(P,Q) = \gcd(6st, 23t^2 - 3s^2) = 1$. Since Q is odd and $U = 6P^2$, $X = 2PQ$ are even, all four $a_i = V \pm U \pm X$ are odd. Any odd prime dividing all four a_i must divide $4V = 4Q^2$ and $4U = 24P^2$, which is impossible since $\gcd(P,Q) = 1$ and $3 \nmid Q$. Therefore, $\gcd(a_1, a_2, a_3, a_4) = 1$.

Similarly, for (b_1, b_2, b_3, b_4), note that $b_1 + b_4 = 46t^2$, $b_2 + b_3 = 6s^2$, and $b_1 - b_4 = 6st$. Since $s \not\equiv t \pmod 2$, either t is odd and s is even (making b_1, b_4 odd) or s is odd and t is even (making b_2, b_3 odd). Thus $2 \nmid \gcd(b_1, b_2, b_3, b_4)$. Any odd prime dividing all four b_j must divide $\gcd(46t^2, 6s^2) = 2\gcd(23t^2, 3s^2) = 2$, which is impossible. Hence $\gcd(b_1, b_2, b_3, b_4) = 1$.

We next establish non-vanishing and pairwise absolute-value distinctness for all $t > s \geq 1$.

- *Non-vanishing and ordering of $|b_j|$:* For $t > s \geq 1$, write $t = s + d$ with $d \geq 1$. Then $|b_3(s,t)| = 3s(t-s) = 3sd \geq 3 > 0$, and the consecutive differences are all strictly positive:

  $$b_2 - |b_3| = 6s^2 > 0, \qquad b_4 - b_2 = 23d^2 + 40ds + 14s^2 > 0, \qquad b_1 - b_4 = 6st > 0.$$

  Hence $0 < |b_3(s,t)| < b_2(s,t) < b_4(s,t) < b_1(s,t)$ for all $t > s \geq 1$.

- *Non-vanishing and pairwise distinctness of $|a_i|$:* Completing the square in Q and P gives

  $$a_1 = (Q+P)^2 + 5P^2, \quad a_2 = (Q-P)^2 - 7P^2, \quad a_3 = (Q-P)^2 + 5P^2, \quad a_4 = (Q+P)^2 - 7P^2.$$

  Since $P = 6st > 0$ and 7 is not a square in $\mathbb{Q}$, all four $a_i(s,t)$ are non-zero for all $(s,t) \in \mathbb{Z}_{>0}^2$. Moreover, the twelve pairwise differences and sums $a_i \pm a_k$ ($1 \leq i < k \leq 4$) are given by

  $$\begin{aligned}
  &a_1 - a_4 = a_3 - a_2 = 2U = 432s^2t^2, && a_1 - a_3 = a_4 - a_2 = 2X = 24st(23t^2 - 3s^2),\\
  &a_1 - a_2 = 24st(23t^2 + 18st - 3s^2), && a_4 - a_3 = 24st(23t^2 - 18st - 3s^2),\\
  &a_1 + a_2 = a_3 + a_4 = 2(23t^2 - 3s^2)^2, && a_1 + a_3 = 2(Q^2 + 6P^2),\\
  &a_2 + a_4 = 2(Q^2 - 6P^2), && a_1 + a_4 = 2(23t^2 - 3s^2)(23t^2 + 12st - 3s^2),\\
  &a_2 + a_3 = 2(23t^2 - 3s^2)(23t^2 - 12st - 3s^2). &&
  \end{aligned}$$

  Since the quadratic factors $23t^2 - 3s^2$, $23t^2 \pm 18st - 3s^2$, $23t^2 \pm 12st - 3s^2$, and $Q^2 - 6P^2$ have non-square discriminants $276s^2$, $600s^2$, $420s^2$, and $24P^2$, none of $a_i \pm a_k$ can vanish over $\mathbb{Z}_{>0}$, so $|a_1|, |a_2|, |a_3|, |a_4|$ are pairwise distinct. In fact, since $a_1 - a_4 = a_3 - a_2 = 432s^2t^2 > 0$ and $a_4 - a_3 = 24st\big(23(t-s)^2 + 2s(14t - 13s)\big) > 0$, we have $a_2(s,t) < a_3(s,t) < a_4(s,t) < a_1(s,t)$ for all $t > s \geq 1$.

- *Cross-family distinctness ($|a_i(s,t)| \neq |b_j(s,t)|$) and explicit polynomial bound:* First, for all $t > s \geq 1$ satisfying the hypotheses, modulo 3 we have $a_i(s,t) \equiv V \equiv (23t^2)^2 \equiv 1 \pmod 3$ (since $3 \nmid t$), whereas $b_2(s,t) = 3s(s+t) \equiv 0 \pmod 3$ and $b_3(s,t) = 3s(s-t) \equiv 0 \pmod 3$; hence $|a_i(s,t)| \notin \{|b_2(s,t)|, |b_3(s,t)|\}$. Modulo t, we have $a_i(s,t) \equiv 9s^4 \pmod t$, so $\gcd(a_i(s,t), t) = \gcd(9s^4, t) = 1$ (since $\gcd(s,t) = 1$ and $3 \nmid t$). Since $t > s \geq 1$ implies $t \geq 2$, t cannot divide $a_i(s,t)$, whereas $t \mid b_1(s,t) = t(23t + 3s)$ and $t \mid b_4(s,t) = t(23t - 3s)$; hence $|a_i(s,t)| \notin \{|b_1(s,t)|, |b_4(s,t)|\}$.

  Furthermore, whenever $10t \geq 11s$, writing $10t = 11s + d$ with $d \geq 0$ yields the exact polynomial identity

$$10^4\big(a_2(s,t) - b_1(s,t)\big) = 529d^4 + 20516d^3s + (257574s^2 - 2300)d^2 + (1071716s^3 - 53600s)d + s^2(274129s^2 - 311300).$$

  For $s \geq 2$ and $d \geq 0$, every coefficient is strictly positive; for $s = 1$, $t > s$ implies $t \geq 2$ and $d = 10t - 11 \geq 9$, giving $10^4(a_2 - b_1) \geq 529(9^4) - 37171 > 0$. (In particular, for $t = 2s + d_0$ with $d_0 \geq 0$, we have $a_2(s,t) - b_1(s,t) = 529d_0^4 + 3956d_0^3s + (10686s^2 - 23)d_0^2 + (12236s^3 - 95s)d_0 + s^2(4921s^2 - 98) > 0$.) Thus $\min_{1\leq i\leq 4} |a_i(s,t)| = a_2(s,t) > b_1(s,t) = \max_{1\leq j\leq 4} |b_j(s,t)|$, establishing (13).

Thus Conditions **(C1)–(C5)** hold for every such pair (s,t). □

3.2 Family B: Low-Coefficient Polynomials ($m_1 = 2, m_2 = 1$)

Setting $m_1 = 2$ and $m_2 = 1$ in (10) gives $6A_b^2 = 16P^2 + 2Q^2$, or $3A_b^2 = 8P^2 + Q^2$. Modulo 4, this forces $Q = 2Q_0$ and $A_b = 2A_0$, reducing the conic to $2P^2 + Q_0^2 = 3A_0^2$. Parametrizing from the rational point $(P, Q_0, A_0) = (-1, -1, 1)$ in parameters (u, w) gives

$$P = 2u^2 - 2uw - w^2, \qquad Q_0 = 2u^2 + 4uw - w^2, \qquad A_0 = 2u^2 + w^2.$$

Theorem 3.3 (Family B for $n = 6$)**.** *For any integers $u, w \in \mathbb{Z}$, define*

$$\begin{aligned} b_1(u,w) &= 6u^2 + 2uw - w^2, & a_1(u,w) &= 56u^4 + 32u^3w - 16uw^3 + 14w^4,\\ b_2(u,w) &= 2u^2 - 6uw + w^2, & a_2(u,w) &= 24u^4 - 96u^3w - 96u^2w^2 + 48uw^3 + 6w^4,\\ b_3(u,w) &= -2u^2 - 2uw + 3w^2, & a_3(u,w) &= 24u^4 + 96u^2w^2 + 6w^4,\\ b_4(u,w) &= 2u^2 + 6uw + w^2, & a_4(u,w) &= 8u^4 + 128u^3w - 64uw^3 + 2w^4. \end{aligned} \tag{14}$$

Then the identity

$$a_1(u,w)^4 - a_2(u,w)^4 + a_3(u,w)^4 - a_4(u,w)^4 = 6\big(b_1(u,w)^4 - b_2(u,w)^4 + b_3(u,w)^4 - b_4(u,w)^4\big)^2$$

holds identically in $\mathbb{Z}[u,w]$. Moreover, whenever $u, w \in \mathbb{Z} \setminus \{0\}$ satisfy $\gcd(u,w) = 1$, w is odd, and $3 \nmid (u+w)$, we have

$$\gcd(b_1, b_2, b_3, b_4) = 1 \qquad \textit{and} \qquad \gcd(a_1, a_2, a_3, a_4) = 2,$$

*so $\gcd(a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4) = 1$, and all eight integers satisfy Conditions **(C1)–(C5)** of Problem 1.1.*

Proof. Here $B_b = 2P$, $C_b = Q = 2Q_0$, and $A_b = 2A_0$, so $b_1 = A_0 + P + Q_0$, $b_2 = A_0 + P - Q_0$, $b_3 = A_0 - P - Q_0$, and $b_4 = A_0 - P + Q_0$, which expand to the quadratic forms in (14). Likewise, $U = 6P^2$, $V = 4Q_0^2$, and $X = 4PQ_0$ give $a_1 = X + U + V$, $a_2 = X + U - V$, $a_3 = -(X - U - V)$, and $a_4 = X - U + V$. Since $2P^2 + Q_0^2 = 3A_0^2$, equation (10) holds identically, establishing the quartic identity.

Proof that $\gcd(b_1, b_2, b_3, b_4) = 1$*:* Observe that $b_1 + b_3 = 2A_0 = 2(2u^2 + w^2)$, $b_2 + b_4 = 2A_0$, $b_1 + b_2 - b_3 - b_4 = 4P$, and $b_1 - b_2 - b_3 + b_4 = 4Q_0$. When w is odd, $b_j \equiv w^2 \equiv 1 \pmod 2$, so all four b_j are odd and $2 \nmid \gcd(b_1, b_2, b_3, b_4)$. Any odd prime $p \mid \gcd(b_1, b_2, b_3, b_4)$ must divide $\gcd(P, Q_0, A_0)$, and hence must divide both $Q_0 - P = 6uw$ and $A_0 = 2u^2 + w^2$ (note also that $\mathrm{Res}_u(P, A_0) = 24w^4$ and $\mathrm{Res}_u(Q_0, A_0) = 48w^4$). If $p > 3$, then $p \mid uw$ and $p \mid (2u^2 + w^2)$, which forces $p \mid u$ and $p \mid w$, contradicting $\gcd(u, w) = 1$. Modulo 3, $A_0 = 2u^2 + w^2 \equiv 2(u - w)(u + w)$ and $P \equiv Q_0 \equiv 2(u + w)^2 \pmod 3$, so $3 \mid \gcd(P, Q_0, A_0)$ if and only if $3 \mid (u + w)$. Since $3 \nmid (u + w)$, we conclude that $\gcd(b_1, b_2, b_3, b_4) = 1$.

Proof that $\gcd(a_1, a_2, a_3, a_4) = 2$*:* Define $a_i' = a_i/2 \in \mathbb{Z}[u, w]$ for $i = 1, 2, 3, 4$, so that

$$a_1' = 3P^2 + 2Q_0^2 + 2PQ_0, \qquad a_2' = 3P^2 - 2Q_0^2 + 2PQ_0,$$
$$a_3' = 3P^2 + 2Q_0^2 - 2PQ_0, \qquad a_4' = -3P^2 + 2Q_0^2 + 2PQ_0.$$

Since w is odd, $P = 2u(u - w) - w^2 \equiv -w^2 \equiv 1 \pmod 2$ is odd, while $2Q_0^2$ and $2PQ_0$ are even. Consequently, every $a_i' \equiv 3P^2 \equiv 1 \pmod 2$ is odd, meaning $2 \mid a_i$ and $4 \nmid a_i$ for each $i = 1, 2, 3, 4$. Let $g' = \gcd(a_1', a_2', a_3', a_4')$; then g' is odd and $\gcd(a_1, a_2, a_3, a_4) = 2g'$. From the linear combinations

$$(a_1' + a_3') + (a_2' - a_4') = 12P^2 \qquad \text{and} \qquad (a_1' + a_3') - (a_2' - a_4') = 8Q_0^2,$$

any odd prime $p \mid g'$ must divide both $3P^2$ and Q_0^2, so $p \mid Q_0$. If $p = 3$, then $3 \mid Q_0 \equiv 2(u + w)^2 \pmod 3$, which contradicts $3 \nmid (u + w)$. If $p > 3$, then $p \mid P$ and $p \mid Q_0$, so p divides both $Q_0 - P = 6uw$ and $2P + Q_0 = 3(2u^2 - w^2)$. Since $p > 3$, $p \mid uw$ and $p \mid (2u^2 - w^2)$, which implies $p \mid u$ and $p \mid w$, contradicting $\gcd(u, w) = 1$. Therefore $g' = 1$ and $\gcd(a_1, a_2, a_3, a_4) = 2$, which implies $\gcd(a_1, \ldots, b_4) = \gcd(2, 1) = 1$.

Verification of Conditions ***(C1)–(C3)****:* Each $b_j(u, w)$ has discriminant $28w^2$ with respect to u, which is not a rational square for $w \neq 0$, so $b_j(u, w) \neq 0$. Similarly, $a_1' + a_3' = 2(3P^2 + 2Q_0^2) > 0$, and $a_i'(u, w) \neq 0$ for each $i = 1, 2, 3, 4$ because the binary quadratic forms $3P^2 \pm 2Q_0^2 \pm 2PQ_0$ in (P, Q_0) either are positive-definite (discriminant $-5Q_0^2 < 0$ for a_1', a_3') or have non-square discriminant $7Q_0^2$ (for a_2', a_4', where $Q_0 \neq 0$ since $\Delta_u(Q_0) = 24w^2$ is not a square); hence $a_i(u, w) \neq 0$.

For pairwise distinctness of the absolute values, we exhibit the explicit factorizations of all twelve b-sums/differences and all twelve a-sums/differences over $\mathbb{Z}[u, w]$. On the b-side:

$$b_1 - b_2 = 2Q_0, \quad b_1 + b_2 = 4u(2u - w), \quad b_1 - b_3 = 4(u + w)(2u - w),$$
$$b_1 + b_3 = 2A_0, \quad b_1 - b_4 = 2P, \quad b_1 + b_4 = 8u(u + w),$$
$$b_2 - b_3 = 2P, \quad b_2 + b_3 = -4w(2u - w), \quad b_2 - b_4 = -12uw,$$
$$b_2 + b_4 = 2A_0, \quad b_3 - b_4 = -2Q_0, \quad b_3 + b_4 = 4w(u + w).$$

On the a-side:

$$a_1 - a_2 = 8Q_0^2, \qquad a_1 + a_2 = 4P(10u^2 + 2uw - 5w^2),$$
$$a_1 - a_3 = 8PQ_0, \qquad a_1 + a_4 = 16(u + w)(2u - w)Q_0,$$
$$a_1 - a_4 = 12P^2, \qquad a_2 - a_3 = -48uwQ_0,$$
$$a_2 + a_3 = 12P^2, \qquad a_2 + a_4 = 8PQ_0,$$
$$a_3 - a_4 = 4P(2u^2 - 14uw - w^2), \qquad a_3 + a_4 = 8Q_0^2,$$

together with the two quartic combinations

$$a_1 + a_3 = 4(20u^4 + 8u^3w + 24u^2w^2 - 4uw^3 + 5w^4),$$
$$a_2 - a_4 = 4(4u^4 - 56u^3w - 24u^2w^2 + 28uw^3 + w^4).$$

The five quadratic factors P, Q_0, A_0, $10u^2 + 2uw - 5w^2$, and $2u^2 - 14uw - w^2$ have discriminants (with respect to u) equal to $12w^2$, $24w^2$, $-8w^2$, $204w^2$, and $204w^2$, none of which is a rational

square for $w \neq 0$. Likewise, the two quartic factors $20u^4 + 8u^3w + 24u^2w^2 - 4uw^3 + 5w^4$ and $4u^4 - 56u^3w - 24u^2w^2 + 28uw^3 + w^4$ are irreducible over $\mathbb{Q}$ and hence have no non-trivial rational zeros. Finally, since $u, w \neq 0$, $3 \nmid (u + w)$ (so $u + w \neq 0$), and w is odd (so $2u - w \neq 0$), none of the linear factors u, w, $u + w$, or $2u - w$ can vanish. Thus none of $b_j \pm b_k$ or $a_i \pm a_k$ vanishes. Finally, since every $a_i(u, w)$ is even whereas every $b_j(u, w)$ is odd, we have $|a_i(u, w)| \neq |b_j(u, w)|$ for all $1 \leq i, j \leq 4$. □

4 Universal Polynomial Parametrization for Any Fixed Positive Integer n

We now solve Problem 1.1 in complete generality when 6 is replaced by an arbitrary fixed positive integer $n \in \mathbb{Z}_{>0}$. The foundation of our construction is a two-stage application of the Universal Quadric Lemma (Lemma 2.2) combined with the following elementary scaling principle.

Lemma 4.1 (Coprime Scaling Principle)**.** *Suppose $(A_1, A_2, A_3, A_4) \in \mathbb{Z}^4$ and $(b_1, b_2, b_3, b_4) \in \mathbb{Z}^4$ are integer 4-tuples satisfying*

$$A_1^4 - A_2^4 + A_3^4 - A_4^4 = n \cdot C^2 \qquad \text{and} \qquad b_1^4 - b_2^4 + b_3^4 - b_4^4 = C \cdot M^2$$

for some non-zero integers $C, M \in \mathbb{Z} \setminus \{0\}$. Define $a_i = MA_i$ for $i = 1, 2, 3, 4$. Then:

(i) The identity $a_1^4 - a_2^4 + a_3^4 - a_4^4 = n \cdot \left(b_1^4 - b_2^4 + b_3^4 - b_4^4\right)^2$ holds.

(ii) If $\gcd(b_1, b_2, b_3, b_4) = 1$, then $\gcd(a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4) = 1$ regardless of M.

Proof. For (i), homogeneity of degree 4 yields

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = M^4\left(A_1^4 - A_2^4 + A_3^4 - A_4^4\right) = M^4\left(nC^2\right) = n\left(CM^2\right)^2 = n\left(b_1^4 - b_2^4 + b_3^4 - b_4^4\right)^2.$$

For (ii), $\gcd(a_1, a_2, a_3, a_4, b_1, b_2, b_3, b_4)$ divides $\gcd(b_1, b_2, b_3, b_4) = 1$, so the joint greatest common divisor is 1. □

We now state and prove the main theorem of this paper, which furnishes an explicit polynomial parametrization in $\mathbb{Z}[n, p, q, s, t]$ valid for every positive integer $n \geq 1$.

Theorem 4.2 (Universal Polynomial Parametrization for Arbitrary $n \in \mathbb{Z}_{>0}$)**.** *Let $n \in \mathbb{Z}_{>0}$ be any fixed positive integer, and let p, q, s, t be integer parameters.*

- ***Stage 1 (Base A-tuple):*** *Set $u_a = 3np^2$ and $r_a = q^2$, and define*

$$\begin{aligned} A_1(p,q) &= (3np^2 + q^2)(3np^2 + 5q^2), & A_2(p,q) &= (3q^2 - 3np^2)(3np^2 + q^2), \\ A_3(p,q) &= -9n^2p^4 - 12np^2q^2 + 3q^4, & A_4(p,q) &= 9n^2p^4 + 5q^4, \end{aligned} \tag{15}$$

  along with the auxiliary polynomial

$$C_0(p,q) = pq\left(3np^2 + q^2\right)\left(9n^2p^4 + 6np^2q^2 + 7q^4\right) \qquad \text{and} \qquad C(p,q) = 12\,C_0(p,q). \tag{16}$$

- ***Stage 2 (Target b-tuple and a-tuple):*** *Set $u_b = C_0(p,q)s^2$ and $r_b = t^2$, and define the polynomials $b_1, b_2, b_3, b_4 \in \mathbb{Z}[n, p, q, s, t]$ (of degree 4 in (s, t)):*

$$\begin{aligned} b_1(s,t) &= (C_0s^2 + t^2)(C_0s^2 + 5t^2), & b_2(s,t) &= (3t^2 - C_0s^2)(C_0s^2 + t^2), \\ b_3(s,t) &= -C_0^2s^4 - 4C_0s^2t^2 + 3t^4, & b_4(s,t) &= C_0^2s^4 + 5t^4, \end{aligned} \tag{17}$$

  the multiplier polynomial $M(s,t)$ (of degree 6 in (s,t)):

$$M(s,t) = 2st\left(C_0s^2 + t^2\right)\left(C_0^2s^4 + 2C_0s^2t^2 + 7t^4\right), \tag{18}$$

  and the polynomials $a_1, a_2, a_3, a_4 \in \mathbb{Z}[n, p, q, s, t]$ (of degree 6 in (s,t)):

$$a_i(s,t) = M(s,t)\,A_i(p,q) \qquad (i = 1, 2, 3, 4). \tag{19}$$

Then:

(i) ***Universal Polynomial Identity:*** *The identity*

$$a_1(s,t)^4 - a_2(s,t)^4 + a_3(s,t)^4 - a_4(s,t)^4 = n \cdot \Big(b_1(s,t)^4 - b_2(s,t)^4 + b_3(s,t)^4 - b_4(s,t)^4\Big)^2$$

holds identically in the polynomial ring $\mathbb{Z}[n,p,q,s,t]$.

(ii) ***Infinitely Many Non-Degenerate Primitive Solutions for Every*** $n \geq 1$***:*** *Fix any positive integer* $n \geq 1$ *and fix* $(p,q) = (2,1)$ *(or any integers* $p > q \geq 1$ *with* p *even,* q *odd, and* $\gcd(3np, q) = 1$*). Then for every pair of positive integers* $(s,t) \in \mathbb{Z}_{>0}^2$ *satisfying*

$$6 \mid s \qquad \text{and} \qquad \gcd(6C_0 s, t) = 1, \tag{20}$$

the resulting integer 4-tuples (a_1,a_2,a_3,a_4) *and* (b_1,b_2,b_3,b_4) *satisfy all five conditions* ***(C1)–(C5)*** *of Problem 1.1: all eight integers are non-zero, have pairwise distinct absolute values, satisfy the primitivity condition* $\gcd(b_1,b_2,b_3,b_4) = 1$ *(and hence* $\gcd(a_1,\ldots,b_4) = 1$*), and satisfy (1).*

Proof. (i) Applying Lemma 2.2(ii) with $u = u_a = 3np^2$ and $r = r_a = q^2$ yields $3r_a u_a = 9np^2q^2 = n(3pq)^2$ and $W_a = u_a^2 + 2u_a r_a + 7r_a^2 = 9n^2p^4 + 6np^2q^2 + 7q^4$. Thus (7) gives

$$A_1^4 - A_2^4 + A_3^4 - A_4^4 = n(3pq)^2 \cdot \big(4(u_a + r_a)W_a\big)^2 = n \cdot \big(12pq(u_a + r_a)W_a\big)^2 = n \cdot C(p,q)^2.$$

Next, applying Lemma 2.2(ii) with $u = u_b = C_0 s^2$ and $r = r_b = t^2$ gives $3r_b u_b = 3C_0 s^2 t^2$ and $W_b = C_0^2 s^4 + 2C_0 s^2 t^2 + 7t^4$. By (7),

$$b_1^4 - b_2^4 + b_3^4 - b_4^4 = 3C_0 s^2 t^2 \cdot \big(4(u_b + r_b)W_b\big)^2 = (12C_0) \cdot \big(2st(u_b + r_b)W_b\big)^2 = C(p,q) \cdot M(s,t)^2.$$

By Lemma 4.1(i), setting $a_i = MA_i$ immediately yields $a_1^4 - a_2^4 + a_3^4 - a_4^4 = n\big(b_1^4 - b_2^4 + b_3^4 - b_4^4\big)^2$ in $\mathbb{Z}[n,p,q,s,t]$.

(ii) Fix $n \geq 1$ and $(p,q) = (2,1)$ (or any $p > q \geq 1$ with $2 \mid p$, $2 \nmid q$, and $\gcd(3np, q) = 1$). Then $u_a = 3np^2 > 3q^2 = 3r_a > 0$ (since $p > q \geq 1$ and $n \geq 1$), $\gcd(u_a, r_a) = 1$, u_a is even while $r_a = q^2$ is odd, and $u_a + r_a = 3np^2 + q^2 \equiv q^2 \equiv 1 \not\equiv 0 \pmod 3$. By Lemma 2.3, $\gcd(A_1,A_2,A_3,A_4) = 1$, and $|A_1|, |A_2|, |A_3|, |A_4|$ are four pairwise distinct positive integers, with $C_0(p,q) \geq C_0(2,1) \geq 4550 > 0$.

Now let $s,t \in \mathbb{Z}_{>0}$ satisfy $6 \mid s$ and $\gcd(6C_0 s, t) = 1$. Then for $u_b = C_0 s^2$ and $r_b = t^2$:

1. $\gcd(u_b, r_b) = \gcd(C_0 s^2, t^2) = 1$ since $\gcd(6C_0 s, t) = 1$.
2. u_b is even (since $6 \mid s$) and $r_b = t^2$ is odd (since $\gcd(6,t) = 1$), so $u_b \not\equiv r_b \pmod 2$.
3. Modulo 3, $u_b = C_0 s^2 \equiv 0 \pmod 3$ (since $3 \mid s$) and $r_b = t^2 \equiv 1 \pmod 3$ (since $3 \nmid t$), so $u_b + r_b \equiv 1 \not\equiv 0 \pmod 3$.

Therefore, by Lemma 2.3, $\gcd(b_1,b_2,b_3,b_4) = 1$, and $|b_1|, |b_2|, |b_3|, |b_4|$ are four pairwise distinct positive integers. By Lemma 4.1(ii), $\gcd(a_1,a_2,a_3,a_4,b_1,b_2,b_3,b_4) = 1$.

Finally, since $|A_1|, |A_2|, |A_3|, |A_4| \geq 1$ are four pairwise distinct positive integers and $M(s,t) > 0$, the four numbers $|a_i| = M|A_i|$ are also pairwise distinct positive integers. Moreover, for any $u_b, r_b > 0$, the explicit algebraic comparisons

$$\begin{aligned} b_1 - |b_2| &= (u_b + r_b)\big((u_b + 5r_b) - |3r_b - u_b|\big) = (u_b + r_b) \cdot \min(2u_b + 2r_b,\, 8r_b) > 0, \\ b_1 - |b_3| &\geq (u_b^2 + 6u_b r_b + 5r_b^2) - (u_b^2 + 4u_b r_b + 3r_b^2) = 2r_b(u_b + r_b) > 0, \\ b_1 - b_4 &= 6u_b r_b > 0 \end{aligned}$$

show that $\max_{1\le j\le 4}|b_j| = b_1 = (u_b + r_b)(u_b + 5r_b)$. Since $s, t, u_b, r_b \ge 1$ implies $2st \ge 2$, we have the explicit positive lower bound

$$2st\left(u_b^2 + 2u_b r_b + 7r_b^2\right) - (u_b + 5r_b) \ge u_b(2u_b - 1) + 4u_b r_b + r_b(14r_b - 5) \ge 1 + 4 + 9 = 14 > 0.$$

Multiplying by $u_b + r_b > 0$ gives

$$\begin{aligned}\max_{1\le j\le 4}|b_j| = (u_b + r_b)(u_b + 5r_b) &< 2st(u_b + r_b)\left(u_b^2 + 2u_b r_b + 7r_b^2\right)\\ &= M(s,t) \le M(s,t)\min_{1\le i\le 4}|A_i| = \min_{1\le i\le 4}|a_i|.\end{aligned}$$

Thus none of $|a_i|$ can equal any $|b_j|$, so all eight integers have pairwise distinct absolute values, completing the proof. □

4.1 Square-Free Reduction and Compact Parametrizations

While Theorem 4.2 works uniformly in $\mathbb{Z}[n, p, q, s, t]$ without factoring C, for any specific integer n one can obtain much smaller coefficients by factoring out the square part of $3C$.

Corollary 4.3 (Square-Free Reduction Method)**.** *Suppose (A_1, A_2, A_3, A_4) is any primitive 4-tuple of non-zero integers with pairwise distinct absolute values satisfying $A_1^4 - A_2^4 + A_3^4 - A_4^4 = nC^2$. Write $3C = d_{\mathrm{sf}} S_0^2$, where $d_{\mathrm{sf}} \in \mathbb{Z}_{>0}$ is square-free and $S_0 \in \mathbb{Z}_{>0}$. Set $u_b = d_{\mathrm{sf}} s^2$ and $r_b = t^2$ in Lemma 2.2. Then for any coprime integers $s, t \ge 1$ such that $\gcd(d_{\mathrm{sf}} s, t) = 1$, $d_{\mathrm{sf}} s^2 + t^2 \equiv \pm 1 \pmod 6$, and $S_0 \mid 12st(u_b + r_b)W_b$, the integer multiplier*

$$M = \frac{12st\left(d_{\mathrm{sf}} s^2 + t^2\right)\left(d_{\mathrm{sf}}^2 s^4 + 2d_{\mathrm{sf}} s^2 t^2 + 7t^4\right)}{S_0}$$

*and the tuples $a_i = MA_i$, $b_j = t_j(u_b, r_b)$ satisfy all conditions **(C1)–(C5)** of Problem 1.1 whenever $\max_j |b_j| < M \min_i |A_i|$. In particular, whenever $C = 3k^2$ for some positive integer k (so $3C = 9k^2$, $d_{\mathrm{sf}} = 1$, and $S_0 = 3k$), one may choose the fixed primitive tuple $(b_1, b_2, b_3, b_4) = (105, 55, 31, 81)$ at $(s, t) = (1, 2)$, which satisfies $105^4 - 55^4 + 31^4 - 81^4 = 70{,}276{,}800 = 3 \cdot 4840^2$, and set $M = 14520/S_0 = 4840/k$ and $a_i = MA_i = (4840/k)A_i$ whenever $k \mid 4840$.*

Proof. Since $3r_b u_b = 3d_{\mathrm{sf}} s^2 t^2 = (9C/S_0^2)s^2 t^2 = C(3st/S_0)^2$, Lemma 2.2(ii) gives

$$b_1^4 - b_2^4 + b_3^4 - b_4^4 = C\left(\frac{12st(u_b + r_b)W_b}{S_0}\right)^2 = CM^2.$$

Applying Lemma 2.3 and Lemma 4.1 completes the proof. □

4.2 Conditional Ternary Conic Criterion for Degree-$(4, 2)$ Families

We also record a conditional ternary conic criterion under which (a_1, a_2, a_3, a_4) and (b_1, b_2, b_3, b_4) share the same quadratic factor $W = 3\alpha^2 + \beta^2 + \gamma^2$ of degree $(4, 2)$ without applying the two-stage scaling multiplier M of Lemma 4.1.

Theorem 4.4 (Conditional Degree-$(4, 2)$ Conic Criterion)**.** *Let $\alpha, \beta, \gamma \in \mathbb{Z}$ be integers such that $\alpha + \beta + \gamma$ is even and $\beta\gamma \ne 0$, and define*

$$X = 2\alpha\gamma, \qquad U = 2\beta\gamma, \qquad V = \gamma^2 - \beta^2 - 3\alpha^2, \qquad W = 3\alpha^2 + \beta^2 + \gamma^2.$$

Set $a_1 = X + U + V$, $a_2 = X + U - V$, $a_3 = X - U - V$, $a_4 = X - U + V$ and $b_1 = (\alpha + \beta + \gamma)/2$, $b_2 = (\alpha + \beta - \gamma)/2$, $b_3 = (\alpha - \beta - \gamma)/2$, $b_4 = (\alpha - \beta + \gamma)/2$. Then $3X^2 + U^2 + V^2 = W^2$ holds identically, and (since $\beta\gamma \ne 0$ and $W > 0$)

$$a_1^4 - a_2^4 + a_3^4 - a_4^4 = n \cdot \left(b_1^4 - b_2^4 + b_3^4 - b_4^4\right)^2 \iff 32\left(\gamma^2 - \beta^2 - 3\alpha^2\right) = n\beta\gamma.$$

In particular, whenever the ternary quadratic form $Q_n(\alpha,\beta,\gamma) := 32\gamma^2 - n\beta\gamma - 32\beta^2 - 96\alpha^2 = 0$ possesses a non-trivial integer solution $(\alpha_0,\beta_0,\gamma_0) \in \mathbb{Z}^3$ with $\beta_0\gamma_0 \neq 0$, defining the homogeneous polynomials $D(s,t) = -32(3s^2+t^2)$ and $L(s,t) = 192\alpha_0 s + (64\beta_0 + n\gamma_0)t$ and setting

$$\alpha(s,t) = 2\big(D(s,t)\alpha_0 + sL(s,t)\big), \quad \beta(s,t) = 2\big(D(s,t)\beta_0 + tL(s,t)\big), \quad \gamma(s,t) = 2\,D(s,t)\gamma_0 \tag{21}$$

produces homogeneous quadratic polynomials $\alpha(s,t),\beta(s,t),\gamma(s,t) \in \mathbb{Z}[s,t]$ of degree 2 with $\alpha+\beta+\gamma$ even that satisfy $Q_n(\alpha(s,t),\beta(s,t),\gamma(s,t)) = 0$ identically, yielding a two-variable homogeneous polynomial parametrization $(a_i(s,t),b_j(s,t)) \in \mathbb{Z}[s,t]$ of degree $(4,2)$ satisfying (1).

Proof. Direct expansion gives

$$3X^2 + U^2 + V^2 = 12\alpha^2\gamma^2 + 4\beta^2\gamma^2 + (\gamma^2 - \beta^2 - 3\alpha^2)^2 = (\gamma^2 + \beta^2 + 3\alpha^2)^2 = W^2.$$

By Lemma 2.1, $a_1^4 - a_2^4 + a_3^4 - a_4^4 = 16UVW^2 = 32\beta\gamma(\gamma^2 - \beta^2 - 3\alpha^2)W^2$, while $b_1^4 - b_2^4 + b_3^4 - b_4^4 = \beta\gamma(3\alpha^2 + \beta^2 + \gamma^2) = \beta\gamma W$. Hence

$$\big(a_1^4 - a_2^4 + a_3^4 - a_4^4\big) - n\big(b_1^4 - b_2^4 + b_3^4 - b_4^4\big)^2 = -\beta\gamma W^2\big(96\alpha^2 + 32\beta^2 + n\beta\gamma - 32\gamma^2\big).$$

Since $\beta\gamma \neq 0$ implies $W = 3\alpha^2 + \beta^2 + \gamma^2 > 0$, we have $\beta\gamma W^2 \neq 0$; dividing by $\beta\gamma W^2$ yields the equivalence with $32(\gamma^2 - \beta^2 - 3\alpha^2) = n\beta\gamma$.

Finally, let $B_n(v,w)$ denote the symmetric bilinear form associated with $Q_n(v) = B_n(v,v)$. For $v_0 = (\alpha_0,\beta_0,\gamma_0)$ and the direction vector $d = (s,t,0)$, we have $Q_n(d) = D(s,t) = -32(3s^2+t^2)$ and $-2B_n(v_0,d) = 192\alpha_0 s + (64\beta_0 + n\gamma_0)t = L(s,t)$. The vector $v(s,t) = 2\big(Q_n(d)v_0 - 2B_n(v_0,d)d\big)$ has components (21), which are homogeneous quadratic polynomials of degree 2 in $\mathbb{Z}[s,t]$ with even coefficients, and satisfies

$$\begin{aligned} Q_n\big(\alpha(s,t),\beta(s,t),\gamma(s,t)\big) &= 4\,Q_n(d)^2 Q_n(\alpha_0,\beta_0,\gamma_0) + 8\,Q_n(d)B_n(v_0,d)\big(-2B_n(v_0,d) + 2B_n(v_0,d)\big) \\ &= 4\,D(s,t)^2 Q_n(v_0) = 0. \end{aligned}$$

Since α,β,γ are homogeneous of degree 2 in (s,t), the linear combinations $b_j = (\alpha \pm \beta \pm \gamma)/2 \in \mathbb{Z}[s,t]$ are homogeneous of degree 2, while X, U, V and $a_i = X \pm U \pm V \in \mathbb{Z}[s,t]$ are homogeneous of degree 4. □

Remark 4.5 (Scope, Arithmetic Existence, and Common Factors in Theorem 4.4). Three important features distinguish Theorem 4.4 from Theorems 3.2, 3.3, and 4.2:

(a) *Conditional existence:* Unlike Theorem 4.2 (which holds unconditionally for every $n \geq 1$), Theorem 4.4 requires the ternary quadric $Q_n(\alpha,\beta,\gamma) = 0$ to possess a rational point with $\beta\gamma \neq 0$, which fails for most small n. For $1 \leq n \leq 12$, a search over $|\alpha|,|\beta|,|\gamma| \leq 60$ finds non-trivial integer zeros of Q_n only for $n = 6$, e.g. $(\alpha_0,\beta_0,\gamma_0) = (1,-4,4)$, and for $n = 9$, e.g. $(\alpha_0,\beta_0,\gamma_0) = (1,-16,14)$.

(b) *Polynomial content and scaling:* Although the parametrization (21) avoids the two-stage degree-$(6,4)$ multiplier $M(s,t)$ of Lemma 4.1, the raw polynomials (21) carry a large fixed numerical content and a joint greatest common divisor at integer specializations. For example, for $n = 6$ with base point $(\alpha_0,\beta_0,\gamma_0) = (1,-4,4)$, the polynomials $\alpha(s,t) = 16(12s^2 - 29st - 4t^2)$, $\beta(s,t) = 16(48s^2 + 24st - 13t^2)$, and $\gamma(s,t) = -256(3s^2+t^2)$ give $b_j(s,t)$ with polynomial content 8 and $a_i(s,t)$ with polynomial content 256, so the raw specialization at every integer pair (s,t) has joint gcd ≥ 8 before dividing out common factors.

(c) *Non-degeneracy:* We omit a general arithmetic analysis of Conditions **(C1)–(C4)** for (21), as the unconditional families in Theorems 3.2, 3.3, and 4.2 already provide complete, primitive, non-degenerate solutions for $n = 6$ and for every $n \geq 1$.

5 Explicit Numerical Examples and Computational Verification

All polynomial identities and numerical entries in Tables 1, 2, and 3 have been computed directly from the exact formulas in Sections 3 and 4 and verified by exact integer arithmetic. Table 1 displays concrete primitive solutions for $n = 6$ obtained from Family A (Theorem 3.2, using admissible pairs (s,t) with $t > s \geq 1$) and Family B (Theorem 3.3). Note that since $(-x)^4 = x^4$, we record the positive absolute values of all terms; every row consists of eight pairwise distinct positive integers with joint gcd $= 1$, and satisfies $a_1^4 - a_2^4 + a_3^4 - a_4^4 = 6S_b^2$ where $S_b = b_1^4 - b_2^4 + b_3^4 - b_4^4$.

Table 2 presents compact primitive solutions across $1 \leq n \leq 10$ constructed via the square-free reduction of Corollary 4.3 with $d_{\text{sf}} = 1$ and $(s,t) = (1,2)$, where $u_b = 1$, $r_b = 4$, and $(b_1, b_2, b_3, b_4) = (105, 55, 31, 81)$ is primitive ($\gcd(105, 55, 31, 81) = 1$) with $105^4 - 55^4 + 31^4 - 81^4 = 70{,}276{,}800 = 3 \cdot 4840^2$. The small primitive base tuples (A_1, A_2, A_3, A_4) in Table 2 were found by a direct computer search over small integer quadruples satisfying $A_1^4 - A_2^4 + A_3^4 - A_4^4 = nC^2$ with $C = 3k^2$ (so $3C = S_0^2$ with $S_0 = 3k$ and $k \mid 4840$), rather than from Stage 1 of Theorem 4.2.

Finally, Table 3 illustrates Stage 1 of the unreduced Universal Polynomial Parametrization (Theorem 4.2) evaluated directly from formula (16) at $(p,q) = (2,1)$ for $1 \leq n \leq 10$, where $C_0(2,1) = 2(12n+1)(144n^2+24n+7)$, showing the primitive base tuples (A_1, A_2, A_3, A_4) and auxiliary values $C_0(2,1)$ and $C(2,1) = 12C_0(2,1)$ that feed into Stage 2 for any (s,t) satisfying (20). To illustrate the full two-stage output of Theorem 4.2 at the smallest admissible Stage 2 parameters $(s,t) = (6,1)$ (which satisfy $6 \mid s$ and $\gcd(6C_0 s, t) = 1$):

- For $n = 1$, $C_0(2,1) = 4{,}550$ and $(A_1, A_2, A_3, A_4) = (221, -117, -189, 149)$ yield the primitive solution

$$\begin{aligned}(a_1, a_2, a_3, a_4) = (&11{,}655{,}292{,}209{,}834{,}236{,}964,\ -6{,}170{,}448{,}816{,}971{,}066{,}628,\\ &-9{,}967{,}648{,}088{,}953{,}261{,}476,\ 7{,}858{,}092{,}937{,}852{,}042{,}116),\\ (b_1, b_2, b_3, b_4) = (&26{,}831{,}422{,}805,\ -26{,}830{,}112{,}397,\ -26{,}831{,}095{,}197,\ 26{,}830{,}440{,}005).\end{aligned}$$

- For $n = 6$, $C_0(2,1) = 778{,}910$ and $(A_1, A_2, A_3, A_4) = (5621, -5037, -5469, 5189)$ yield the 28-digit/15-digit primitive solution

$$\begin{aligned}(a_1, a_2, a_3, a_4) = (&1{,}487{,}182{,}336{,}948{,}402{,}179{,}367{,}341{,}444,\ -1{,}332{,}669{,}886{,}356{,}360{,}394{,}498{,}007{,}268,\\ &-1{,}446{,}966{,}767{,}616{,}226{,}920{,}291{,}761{,}316,\ 1{,}372{,}885{,}455{,}688{,}535{,}653{,}573{,}587{,}396),\\ (b_1, b_2, b_3, b_4) = (&786{,}284{,}389{,}622{,}165,\ -786{,}284{,}165{,}296{,}077,\\ &-786{,}284{,}333{,}540{,}637,\ 786{,}284{,}221{,}377{,}605).\end{aligned}$$

Table 1: Explicit primitive solutions to $a_1^4 - a_2^4 + a_3^4 - a_4^4 = 6(b_1^4 - b_2^4 + b_3^4 - b_4^4)^2$ from Theorems 3.2 and 3.3.

Family & Parameters	**4-Tuple** $(\lvert a_1\rvert, \lvert a_2\rvert, \lvert a_3\rvert, \lvert a_4\rvert)$	**4-Tuple** $(\lvert b_1\rvert, \lvert b_2\rvert, \lvert b_3\rvert, \lvert b_4\rvert)$	$S_b = b_1^4 - b_2^4 + b_3^4 - b_4^4$	$\gcd(a_i), \gcd(b_j)$
Family B: $(u,w) = (1,1)$	$(86, 114, 126, 74)$	$(7, 3, 1, 9)$	$-4{,}240$	2, 1
Family B: $(u,w) = (2,3)$	$(1934, 2298, 4326, 94)$	$(27, 19, 7, 53)$	$-7{,}486{,}960$	2, 1
Family B: $(u,w) = (3,1)$	$(5366, 1362, 2814, 3914)$	$(59, 1, 21, 37)$	$10{,}437{,}680$	2, 1
Family A: $(s,t) = (1,2)$	$(10921, 4921, 6649, 9193)$	$(98, 9, 3, 86)$	$37{,}529{,}520$	1, 1
Family A: $(s,t) = (1,4)$	$(154201, 112249, 119161, 147289)$	$(380, 15, 9, 356)$	$4{,}789{,}302{,}240$	1, 1
Family A: $(s,t) = (2,5)$	$(406129, 227809, 271009, 362929)$	$(605, 42, 18, 545)$	$45{,}747{,}443{,}280$	1, 1
Family A: $(s,t) = (3,4)$	$(196489, 36073, 98281, 134281)$	$(404, 63, 9, 332)$	$14{,}474{,}386{,}080$	1, 1
Family A: $(s,t) = (4,5)$	$(490609, 64849, 237649, 317809)$	$(635, 108, 12, 515)$	$92{,}110{,}071{,}840$	1, 1

Acknowledgement

This paper is written with assistance of AI.

Table 2: Compact primitive solutions to $a_1^4 - a_2^4 + a_3^4 - a_4^4 = n(b_1^4 - b_2^4 + b_3^4 - b_4^4)^2$ across $1 \le n \le 10$ via Corollary 4.3 with $(b_1, b_2, b_3, b_4) = (105, 55, 31, 81)$ (using computer-searched base tuples (A_1, A_2, A_3, A_4)).

n	**Base Tuple** (A_1, A_2, A_3, A_4)	C	**Multiplier** M	**Solution Tuple** $(a_1, a_2, a_3, a_4) = M \cdot A$	**Joint** gcd
1	$(10, 1, 13, 14)$	12	2,420	$(24200, 2420, 31460, 33880)$	1
2	$(27, 22, 41, 42)$	75	968	$(26136, 21296, 39688, 40656)$	1
3	$(39, 30, 8, 35)$	48	1,210	$(47190, 36300, 9680, 42350)$	1
4	$(8, 3, 9, 10)$	12	2,420	$(19360, 7260, 21780, 24200)$	1
5	$(18, 12, 1, 17)$	12	2,420	$(43560, 29040, 2420, 41140)$	1
6	$(4, 1, 5, 2)$	12	2,420	$(9680, 2420, 12100, 4840)$	1
7	$(25, 11, 30, 33)$	3	4,840	$(121000, 53240, 145200, 159720)$	1
8	$(16, 2, 33, 27)$	300	484	$(7744, 968, 15972, 13068)$	1
9	$(7, 20, 28, 26)$	3	4,840	$(33880, 96800, 135520, 125840)$	1
10	$(8, 4, 1, 7)$	12	2,420	$(19360, 9680, 2420, 16940)$	1

Table 3: Stage 1 primitive base tuples (A_1, A_2, A_3, A_4), $C_0(2,1) = 2(12n+1)(144n^2+24n+7)$, and $C(2,1) = 12C_0(2,1)$ from Theorem 4.2 at $(p, q) = (2, 1)$ for $1 \le n \le 10$, where $A_1^4 - A_2^4 + A_3^4 - A_4^4 = n(12C_0)^2$.

n	**Primitive Base Tuple** (A_1, A_2, A_3, A_4) **at** $(p, q) = (2, 1)$	$u_a = 12n$	$C_0(2,1)$	$C(2,1) = 12C_0(2,1)$	$\gcd(A_1, A_2, A_3, A_4)$
1	$(221, -117, -189, 149)$	12	4,550	54,600	1
2	$(725, -525, -669, 581)$	24	31,550	378,600	1
3	$(1517, -1221, -1437, 1301)$	36	101,750	1,221,000	1
4	$(2597, -2205, -2493, 2309)$	48	235,886	2,830,632	1
5	$(3965, -3477, -3837, 3605)$	60	454,694	5,456,328	1
6	$(5621, -5037, -5469, 5189)$	72	778,910	9,346,920	1
7	$(7565, -6885, -7389, 7061)$	84	1,229,270	14,751,240	1
8	$(9797, -9021, -9597, 9221)$	96	1,826,510	21,918,120	1
9	$(12317, -11445, -12093, 11669)$	108	2,591,366	31,096,392	1
10	$(15125, -14157, -14877, 14405)$	120	3,544,574	42,534,888	1

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